

Small Commercial Building Energy Analysis Dashboard

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30-day load profile analysis with a 30 kW rooftop solar PV scenario

This dashboard shows how much electricity a small commercial building uses over a 30-day (monthly) period. The original building demand was compared with a simulated 30 kW rooftop solar system. In this project, the results obtained show that the solar generation can significantly reduce the amount of electricity imported from the grid during daytime hours. Most of the solar energy is found to have been directly used by the building, while excess solar is exported back to the grid when generation is higher than demand.

Project assumptions have been listed at the bottom of this dashboard

How to read these results:

- 1) kW measures power at a specific moment in time, similar to how fast electricity is being used.
- 2) kWh measures total energy used over time, this is what electricity bills are mainly based on.
- 3) Grid import means electricity brought in from the grid.
- 4) Solar self-consumed means solar energy used directly by the building.
- 5) Exported solar means extra solar energy sent back to the grid.

Key Metrics:	Value	Unit
Total Energy Used	8130.00	kWh
Peak Demand	34.90	kW
Estimated Bill before Solar	2731.20	\$
Estimated Emissions	5691.00	kg CO2
Overnight Base Load	2.39	kW

Figure 1:

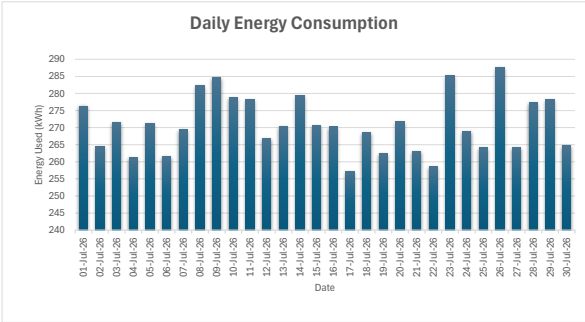


Figure 1 shows how much electricity the building used each day during the 30-day period. Higher bars indicate days with greater total energy use.

Solar Scenario	Value	Unit
Solar PV Size	30.00	kW
Grid Import After Solar	2548.65	kWh
Solar Self-Consumption	81.29	%
Monthly Solar Benefit	1626.99	\$
Simple Payback	1.84	years

Figure 2:

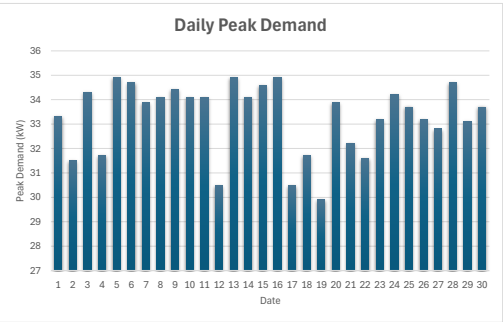


Figure 2 shows the higher power demand reached each day for 30 days. This is important as commercial sites may be charged based on peak demand (the amount of energy used during the peak hour)

Figure 3:

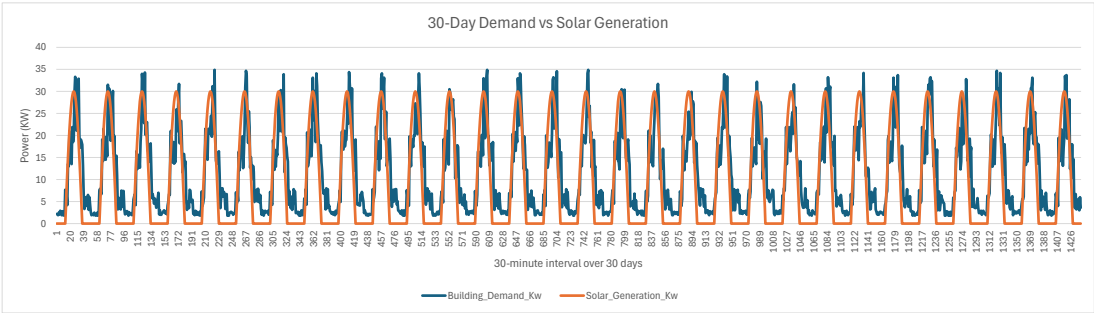
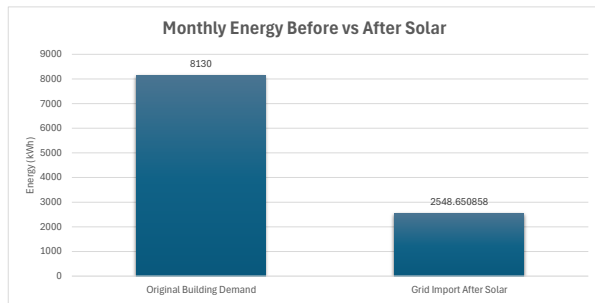


Figure 3 compares the building's electricity demand with the simulated solar generation profile. Solar generation rises during the day and falls to zero overnight as expected.

Figure 4:



This figure 4 shows that the 30 kW solar scenario reduces grid electricity import from 8130 kWh to approximately 2549 kWh.

Figure 5:

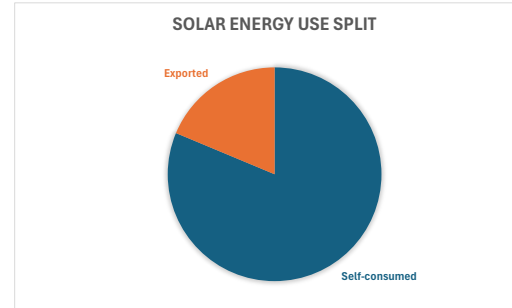


Figure 5 shows how much generated solar energy was used directly by the building compared with how much was exported back to the grid.

Key Insights:

- 1) The building used 8130 kWh over the 30-day period, with demand mainly increasing during business hours (midday).
- 2) The 30 kW rooftop solar scenario reduced estimated grid import from 8130 kWh to approximately 2549 kWh, showing strong alignment between solar generation and daytime building demand.
- 3) Around 81.29% of generated solar energy was used directly by the building, while the remaining solar energy was exported to the grid.
- 4) The estimated monthly solar benefit was approximately \$1627, giving a simple payback of around 1.84 years under the assumed costs and tariffs.
- 5) The overnight base load of 2.39 kW suggests some equipment remains active outside operating hours, as expected for a commercial business.

Project Assumptions:

- 1) The load data is simulated for a small commercial building over a 30-day period.
- 2) Electricity demand is recorded in 30-minute intervals.
- 3) Energy is calculated using: $\text{Energy (kWh)} = \text{Demand (kW)} * 0.5 \text{ hours}$
- 4) A 30 Kw rooftop solar PV system is modelled using a simplified daytime generation curve.
- 5) Solar generation is assumed to peak around midday and fall to 0 overnight.
- 6) Electricity tariff, demand charge, supply charge and feed-in tariff values are all assumed for analysis.
- 7) The analysis does not include detailed electrical design, weather variation, panel orientation, grid connection approval or roof structural assessment.